

Answers learning task 11

2026 – 2027

Exercise 7.21: Let $X_1, X_2, \dots, X_7$ denote a random sample from a population having mean μ and variance σ^2 . Consider the following estimators of μ :

$$\hat{\Theta}_1 = \frac{X_1 + X_2 + \dots + X_7}{7}$$

$$\hat{\Theta}_2 = \frac{2X_1 - X_6 + X_4}{2}$$

- (a) Is either estimator unbiased?
- (b) Which estimator is better? In what sense is it better? Calculate the relative efficiency of the two estimators.

Answer

- (a) Both estimators are unbiased because $E(\hat{\Theta}_1) = \mu$ and $E(\hat{\Theta}_2) = \mu$.
- (b) $\hat{\Theta}_1$ is better because it is more precise. Since $V(\hat{\Theta}_1) = \frac{\sigma^2}{7}$ and $V(\hat{\Theta}_2) = \frac{3\sigma^2}{2}$, the relative efficiency of $\hat{\Theta}_2$ to $\hat{\Theta}_1$ is $\frac{2}{21}$.

Exercise 7.23: Suppose that $\hat{\Theta}_1$ and $\hat{\Theta}_2$ are estimators of the parameter θ . We know that $E(\hat{\Theta}_1) = \theta$, $E(\hat{\Theta}_2) = \theta/2$, $V(\hat{\Theta}_1) = 10$, $V(\hat{\Theta}_2) = 5$. Which estimator is better? In what sense is it better?

Answer

$\hat{\Theta}_1$ is preferred. $\hat{\Theta}_1$ is unbiased with $V(\hat{\Theta}_1) = 10$. $\hat{\Theta}_2$ is biased, but $2\hat{\Theta}_2$ is unbiased with $V(2\hat{\Theta}_2) = 20$. Since $\hat{\Theta}_1$ has a smaller variance among the unbiased options, it is the better estimator.

Exercise 7.25: Let three random samples of sizes $n_1 = 8$, $n_2 = 14$, and $n_3 = 6$ be taken from a population with mean μ and variance σ^2 . Let S_1^2 , S_2^2 , and S_3^2 be the sample variances. Show that $S^2 = (7S_1^2 + 13S_2^2 + 5S_3^2)/25$ is an unbiased estimator of σ^2 .

Answer

Since $E(S_1^2) = \sigma^2$, $E(S_2^2) = \sigma^2$, and $E(S_3^2) = \sigma^2$, we have:

$$E(S^2) = \frac{7}{25}\sigma^2 + \frac{13}{25}\sigma^2 + \frac{5}{25}\sigma^2 = \sigma^2$$

Thus, S^2 is an unbiased estimator of σ^2 .

Exercise 7.29: Data on the oxide thickness of semiconductor wafers are as follows: 425, 431,

416, 420, 421, 436, 418, 410, 431, 433, 423, 426, 411, 435, 436, 428, 411, 426, 407, 437, 422, 428, 413, 416.

- Calculate a point estimate of the mean oxide thickness for all wafers in the population.
- Calculate a point estimate of the standard deviation of oxide thickness for all wafers in the population.
- Calculate the standard error of the point estimate from part (a).
- Calculate a point estimate of the median oxide thickness for all wafers in the population.
- Calculate a point estimate of the proportion of wafers in the population that have oxide thickness of more than 430 angstroms.

Answer

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|--------------|------------|------------|
| (a) 423.3333 | (b) 9.1493 | (c) 1.8676 |
| (d) 424 | (e) 0.2917 | |

Exercise 7.32: Two different plasma etchers in a semiconductor factory have the same mean etch rate μ . However, machine 1 is newer than machine 2 and consequently has smaller variability in etch rate. We know that the variance of etch rate for machine 1 is σ_1^2 , and for machine 2, it is $\sigma_2^2 = a\sigma_1^2$. Suppose that we have n_1 independent observations on etch rate from machine 1 and n_2 independent observations on etch rate from machine 2.

- Show that $\hat{\mu} = \alpha\bar{X}_1 + (1 - \alpha)\bar{X}_2$ is an unbiased estimator of μ for any value of α between zero and one.
- Find the standard error of the point estimate of μ in part (a).
- What value of α would minimize the standard error of the point estimate of μ ?
- Suppose that $a = 4$ and $n_1 = 0.5n_2$. What value of α would you select to minimize the standard error of the point estimate of μ ? How “bad” would it be to arbitrarily choose $\alpha = 0.5$ in this case?

Answer

- $E(\hat{\mu}) = \alpha E(\bar{X}_1) + (1 - \alpha)E(\bar{X}_2) = \alpha\mu + (1 - \alpha)\mu = \mu$
- S.E. = $\sigma_1 \sqrt{\frac{\alpha^2}{n_1} + \frac{a(1-\alpha)^2}{n_2}}$
- $\alpha = \frac{an_1}{n_2 + an_1}$
- $\alpha = \frac{2}{3}$, choosing $\alpha = 0.5$ increases the standard error and is thus not preferable.